

MD ALI AHNAF ABID MAYUKH

Address: Elephant Road, Dhaka, Bangladesh • Phone: +8801408199798 • Email: ahnafabid2@iut-dhaka.edu

[LinkedIn](#) | [Google Scholar](#) | [Website](#)

RESEARCH INTEREST

GIS & Remote Sensing | Ecohydrological modeling | Flood Risk Analysis | Environmental Monitoring | Climate Change | GeoAI

EDUCATION

- **B.Sc. in Civil Engineering** **Jun 2021 - Oct 2025**
Islamic University of Technology (IUT) - OIC, Gazipur, Bangladesh
Concentration: *Geoenvironmental Engineering*, **CGPA:** 3.09/4.00 (181 Credits) [Transcripts](#)
Thesis: “Experimental Study on Seepage Control in Sand Embankments Stabilized with Sodium Lignosulfonate and Supplementary Polymers” - Supervised by Prof. Dr. Hossain Md. Shahin.

TEACHING AND RESEARCH EXPERIENCE

Research Assistant (RA) - [North South University](#), Dhaka, Bangladesh **Feb 2026 - Present**

- Collaborated with faculty at the Department of Civil and Environmental Engineering to develop course materials for the Advanced Foundation Analysis course, applying FEM-based numerical modeling to analyze soil–structure interaction and evaluate foundation performance under complex loading scenarios.
- Contributed to a review paper using a PRISMA-ScR-based scoping review on urban environmental infrastructure governance and a book chapter on sustainable construction materials, conducting large-scale literature synthesis (500+ sources) and supporting technical writing and content development.

Instructor - [LEAD Academy](#), Dhaka, BD **Jan 2025 - Present**

- Developed an AutoCAD course with 13 learning modules with 87 video lectures (9+ hours) with 30 quizzes and 4 final projects.
[Course Link](#) (Published on November, 2025).

AutoCAD Instructor - [Caturjo Architecture Academy](#), Chattogram, Bangladesh **Apr 2025 - Present**

- Delivered a structured 10-week (10+ hours) AutoCAD training program for industry professionals on civil engineering drafting practices in compliance with BNBC 2020 standards and provided learning materials accordingly.
- Trained 20 polytechnic graduate students, conducted live Q&A sessions and webinars.

Chief Instructor: AutoCAD - [IUT CAD Society](#), Gazipur, Bangladesh **Nov 2023 - Oct 2025**

- Taken live classes of a batch of over 30 undergrad students from various departments both online and offline.
- Made the question paper and managed 104 participants as the Head Organizer of the Drafting Contest in the Cennovation 2025.

KEY PUBLICATIONS

Journal Publications:

1. Chowdhury, M. S., **Mayukh, A. A.** & Kim, Y. J. (2025). Spatiotemporal Shoreline Change Assessment And Machine Learning Projections For Coastal Louisiana. Geodata and AI, Article 100089. [DOI](#) [In press]
2. Azad, S. B., Chowdhury, M. I., Sinha, R., & **Mayukh, A. A.** (2024). Improvement Of Engineering Properties Of Different Materials Using Microbial Treatment - A Review. Journal of Harbin Engineering University, 45(1), Article 01. 84-92. [DOI](#)

Conference Papers:

1. **Mayukh, A. A.**, Chowdhury, M. S. GIS-Based Multi-Temporal Land Cover Dynamics in Terrell County, Texas: Assessing Oil-Driven Desertification and Rangeland Resilience (2001–2021). *ASCE 2027 Conference*. [Accepted for presentation]
2. **Mayukh, A. A.**, Chowdhury, M. S. Hybrid Hydro-Geomorphic AHP Framework for Flash Flood Susceptibility: A Case Study of the Limestone Karst Terrain in Kerr County, Texas. *ASCE 2027 Conference*. [Accepted for presentation]
3. Chowdhury, M. S., **Mayukh, A. A.** Spatiotemporal Land Cover Change from 2003 to 2023 and 2033 Projections for Topography Derived Inundation Susceptibility in El Paso County, Texas. *ASCE 2027*. [Accepted for presentation]
4. **Mayukh, A. A.**, Chowdhury, M. S., Kim, Y. J. & Youn, S. (2025). Flood Susceptibility Mapping In San Antonio, Texas Using Multi-Criteria AHP-GIS Integration. *International Geoscience and Remote Sensing Symposium (IGARSS 2026)*. [Accepted for presentation]
5. Prantor, R., **Mayukh, A. A.**, Abid, R. N., Rimi, N. N., Sharaf, K., Fariha, Z., & Joy, R. (2025). Assessment Of Flood Susceptibility In Bangladesh: Integrating SPI Into A GIS-AHP Decision Framework. *International Conference on Civil Engineering Research & Innovations 2025 (ICCEI 2025)* (pp. 1370-1375) [DOI](#)
6. Nishat, M. H., Labib, A., Khan, N. M., Muntaha, S., & **Mayukh, A. A.** (2025). Challenges And Opportunities For Implementing Rooftop Solar In Dhaka’s Residential Sector. *International Conference on Engineering Research, Innovation, and Education 2025 (ICERIE 2025)* (pp. 239-245). Atlantis Press. [DOI](#)
7. **Mayukh, A. A.**, & Labib, A. (2025). A Decadal GIS-Based Analysis Of Turbidity Dynamics And Urban Expansion In The Buriganga River (2013–2023). *Poster Presentation Segment, Cennovation 2025*, Dhaka, Bangladesh. [LINK](#)

SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, ArcMap, Google Earth Engine
- **Engineering & Design :** AutoCAD 2D & 3D, ETABS, PLAXIS 2D & 3D
- **Hydrological Modelling:** HEC-RAS, EPANET, eTank, SWMM
- **Programming Languages:** 1. Python (Advanced) 2. C++ (Intermediate) 3. JavaScript (Basic)

PROFESSIONAL EXPERIENCE

Outside Plant Engineer - Civil Infrastructure (Remote) - [SKARION Eng. Services Ltd.](#), Virginia, USA

Oct 2025 - Present

- Designed 400,000+ ft (500+ residential fiber deployments) of XGS-PON fiber networks, HLD/LLD packages and detailed BOMs.
- Delivered end-to-end OSP designs (aerial & underground) using ArcGIS Pro and AutoCAD, ensuring full compliance with NESC/NEC standards across projects in Williamson County, TX and North Carolina markets (Holly Springs, Greensboro).
- Mentored 25+ junior designers, cutting onboarding time by ~50% and improving overall team delivery efficiency.

INTERNSHIP EXPERIENCE

Industrial Trainee - [DOHWA Engineering Co. Ltd.](#), Dhaka, Bangladesh

Jun 2024 - Jul 2024

- Participated in 6 different heritage site restoration projects and construction site inspections inside Dhaka.
- Attended 2 interim workshops on coastal, water resource, and transportation engineering at Bangladesh Inland Water Transport Corporation (BIWTC) Headquarters and Roads and Highways Department (RHD).
- Used GIS for road network mapping and rural infrastructure planning through the 'Western Economic Corridor & Regional Enhancement Program (WeCARE) Phase-I: Rural Connectivity, Market and Logistic Infrastructure Improvement Project'
- Used AutoCAD LISP for soil cut-and-fill calculations and Excel for engineering data analysis.

EXTRACURRICULAR ACTIVITIES

- **Expert Reviewer - Second Order Draft Review, Special Report on Climate Change and Cities (SRCITIES), Intergovernmental Panel on Climate Change (IPCC - UN)**, May 2026 - Present
Responsible for evaluating the report's scientific and socio-economic accuracy to support a policy-relevant global climate assessment.
- **Reviewer - European Journal of Engineering Research and Reviews (EISSN 3041-4822)**, Jan 2025 - Present
Reviewed 7+ academic manuscripts, providing constructive feedback on engineering research quality and methodology.
- **Student Member - Institute of Electrical and Electronics Engineers (IEEE)**, Dec 2025 - Present
Participated in the IGARSS 2026 Conference organized by IEEE Geoscience and Remote Sensing Society (GRSS).
- **Head of Public Relations - IUT Photographic Society**, Aug 2022 - Oct 2025
Raised BDT 25,000 in corporate sponsorship for 'Break The Circle XI' - international photography competition.
- **Program Executive - Space and Environment Research Center (SERC)**, Jan 2025 - Oct 2025
Project No. SP-9/24 - "Developing Space-Based Integrated Disaster Management System in Asia and the Pacific"
- **Head of Creative Fields - IUT Supply Chain Alliance**, Jan 2024 - Jun 2025
Directed design of 10+ event campaigns with a 5-member design team.

TEST SCORES

- **GRE:** 312 (Quant-161, Verbal-151, AWA- 3.5) – 22 February, 2026 [Test Report Form](#)
- **IELTS:** Overall Band 7.5 (Listening-7.0, Reading-7.0, Writing-7.5, Speaking-8.0) – 25 May 2025 [Test Report Form](#)

PROJECTS

- **Analysis of Shoreline Dynamics at Cameron Beach, Louisiana (2014 - 2023) and prediction using machine learning (2023 - 2033).**
Used GEE, DSAS plugin, ArcGIS Pro and XGBoost model to analyze and predict shoreline change from shorelines derived from NDWI.
- **Developed and evaluated a multi-method cloud masking and compositing framework for Sentinel-2 imagery.**
Implemented rule-based, machine learning, and deep learning cloud-masking workflows in GEE and Python for Sentinel-2 imagery.
- **Performed GIS-based multi-temporal analysis of land cover dynamics in Terrell County, Texas (2001–2021)**
Integrated NLCD, Landsat NDVI, and oil well data to analyze desertification, vegetation recovery, and rangeland resilience.
- **Flood Susceptibility Mapping of Kerr County, Texas using AHP-driven Multi-Criteria GIS Analysis.**
Developed a flood susceptibility map for Kerr County using AHP-MCDA and ROC validation with 9 geomorphological factors.
- **UAV-Based Digital Topographic Survey of RBD Eco Resort, Sreepur, Gazipur for ANTS Drone Hub.**
Surveyed drone imagery with boundary, area, and contour mapping for resort land development using ArcGIS Pro and AutoCAD.
- **LiDAR-derived Digital Elevation Model (DEM) Generation for Plaquemines Parish, Louisiana (2024).**
Processed LiDAR point clouds in ArcGIS Pro, filtered vegetation, and generated DEM in ArcGIS Pro.
- **Scientific Poster Presentation Turbidity Mapping and Urban Encroachment Analysis of the Buriganga River (2013–2023)**
Mapped turbidity and LULC in GEE with 2 km buffer analysis to quantify urban expansion impacts on the Buriganga River, Dhaka.
- **Design and Settlement Analysis of Mat and Isolated Footings of a Residential Building in Brahmanbaria.**
Modeled mat and isolated footings of a 12-story residential building using ETABS and PLAXIS 3D with site-specific soil parameters.
- **Urban Stormwater Modeling of Azimpur Colony, Dhaka using SWMM.**
Conducted a detailed hydrologic and hydraulic analysis of the Azimpur Colony catchment in Dhaka using EPA SWMM 5.2.